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MetaMARL: A Framework for Mechanism Design in Multi-Agent Reinforcement Systems

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Summary

Many societal and environmental systems involve multiple adaptive decision-makers whose individual objectives conflict with long-term collective outcomes. Whether that conflict is resolved depends on the rules under which those decision-makers behave (Dietz et al., 2003; Ostrom, 1990). Examples include common-pool resources such as fisheries, forests, and water systems (Koster et al., 2025), as well as public policies such as taxation and emissions regulation (Koster et al., 2022; Zheng et al., 2022). Experimenting with new rules to govern these systems requires simultaneously modeling strategic agent behavior at the local level and the regulatory mechanisms where these agents are embedded at the global level. MetaMARL additionally supports the composition of heterogeneous agent behaviors and regulatory mechanisms through reusable agent and mechanism objects. Its residual state representation enables multiple interventions to be combined within shared environment dynamics, providing a foundation for studying both bilevel optimization and more general multi-agent interactions.

`MetaMARL` is an open-source Python framework for mechanism design in multi-agent reinforcement learning (MARL). The framework supports mechanism design as a bilevel optimization problem in which an inner optimization trains adaptive agents using reinforcement learning, while an outer optimization searches for environment-level mechanisms that optimize system-wide objectives. It also supports multi-agent experiments without an outer optimization loop.

The framework includes a fishery benchmark and supports the development of more realistic water-reservoir management environments, while also providing modular abstractions for environments, mechanism spaces, optimization algorithms, and experiment management. These components enable researchers to develop new benchmark environments, implement custom regulatory mechanisms, and evaluate existing or novel mechanism design algorithms within a unified experimental framework. Although motivated by renewable resource management, `MetaMARL` is designed to support a broad class of adaptive governance problems in which institutional rules themselves become optimization variables.

Statement of need

Most reinforcement learning software focuses on learning optimal agent policies within fixed environments. Libraries such as RLlib (Liang et al., 2018) and Stable-Baselines3 (Raffin et al., 2021) provide scalable implementations of modern reinforcement learning algorithms but assume that the environment dynamics, reward functions, and governing rules are predefined and fixed.

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Many real-world decision problems instead require optimizing the rules of the environment rather than only the behavior of its participants. Examples include resource quotas, taxation schemes, fines, enforcement policies, and other institutional mechanisms that influence the incentives faced by learning agents. These problems naturally form bilevel optimization problems in which adaptive agents respond to regulatory interventions while regulators seek mechanisms that optimize long-term social or ecological objectives.

Developing research software for these settings presents several challenges:

- integrating reinforcement learning with mechanism optimization,
- supporting non-stationary environments induced by adaptive agents,
- separating mechanism definitions from environment implementations,
- enabling reproducible experiments across different optimization algorithms.

More realistic benchmarks are also needed to evaluate these methods beyond stylized environments, particularly when agents interact through shared physical resources and face competing objectives.

`MetaMARL` addresses these challenges through a modular architecture that decouples environments, mechanism spaces, reinforcement learning algorithms, and outer-loop optimization methods. The framework enables researchers to rapidly prototype new regulatory mechanisms while reusing existing learning algorithms and experimental infrastructure.

State of the field

Several mature reinforcement learning frameworks provide scalable implementations of modern policy optimization algorithms. RLlib (Liang et al., 2018) offers distributed reinforcement learning and extensive support for multi-agent training, while Stable-Baselines3 (Raffin et al., 2021) provides a lightweight interface for many standard algorithms. PettingZoo (Terry et al., 2021) standardizes multi-agent environments and OpenSpiel (Lanctot & others, 2019) provides a broad collection of games for reinforcement learning and game-theoretic research.

These libraries primarily focus on learning agent policies under fixed environmental rules. They provide limited support for optimizing the mechanisms that define those environments.

`MetaMARL` complements rather than replaces existing reinforcement learning libraries. It builds upon RLlib for scalable policy optimization while introducing abstractions for parameterized mechanism spaces, regulator environments, bilevel optimization workflows, and experiment reporting. This allows researchers to study adaptive governance problems where both policies and institutional rules evolve together.

Related work in computational mechanism design includes the AI Economist (Zheng et al., 2022), which investigates the joint adaptation of economic agents and a regulatory planner through two-level reinforcement learning. MetaMARL addresses a complementary software-engineering objective: providing reusable abstractions for agents, mechanisms, environment dynamics, and interchangeable optimization interfaces across different application domains. Rather than prescribing a particular economic simulation or regulatory learning algorithm, the framework emphasizes compositional experiment construction and the systematic evaluation of alternative mechanism designs.

Software design

The framework follows a modular architecture composed of four primary components.

Environment layer. Environment classes define system dynamics independently from regulatory mechanisms, enabling different application domains, from simple games to realistic resource-management models, to reuse common optimization infrastructure.

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Mechanism layer. Mechanism spaces describe parameterized regulatory policies such as quotas, fines, harvest restrictions, or other intervention strategies. New mechanisms can be introduced without modifying optimization algorithms.

Optimization layer. The framework separates inner-loop policy optimization from outer-loop mechanism optimization. Reinforcement learning algorithms such as PPO and APPO are used for agent learning through RLlib, while black-box optimization algorithms such as Evolution Strategies optimize mechanism parameters.

Experiment infrastructure. Configuration objects define complete experiments in a reproducible manner, including environments, optimization algorithms, reporting backends, and training settings. This modular design allows researchers to exchange optimization algorithms or environments with minimal code changes.

Together, these abstractions allow new research problems to be expressed by combining reusable software components rather than implementing specialized training pipelines for each application.

Composable residual mechanisms. A central design choice in MetaMARL is to represent the effects of agents and mechanisms as structured residuals over a shared Markov decision process state. Each mechanism independently maps actions and environmental information to state-update contributions. These residuals are composed through vectorized operations: additive quantities are accumulated, while applicable constraints are combined by intersection. This separates the definition of individual interventions from their composition, allowing users to construct complex regulatory environments from reusable mechanisms. The abstraction supports heterogeneous agents, configurable leader–follower interactions, and extensions involving additional learning or optimization algorithms.

Research impact statement

`MetaMARL` has been developed as research infrastructure for investigating mechanism design in adaptive multi-agent systems. Its common-pool fishery benchmark provides an executable research workflow in which learning agents interact with a renewable resource under configurable regulatory interventions. Researchers can compare mechanism configurations, train agent policies using PPO or APPO, and evaluate the resulting behavior using economic, ecological, and optimization metrics.

The benchmark demonstrates the framework's capacity to connect domain-specific environment dynamics with reusable mechanism and optimization components. The accompanying configuration files, tests, and tutorials provide a reproducible starting point for extending these experiments to new mechanisms, agent populations, and evaluation objectives.

A central objective of MetaMARL is to support the development of increasingly realistic benchmarking environments for studying mechanism design in multi-agent systems, moving beyond the stylized settings commonly used in the field. Such environments provide testbeds for investigating social dilemmas, multi-agent alignment, and AI safety, where collective outcomes emerge from interactions among adaptive agents under shared environmental constraints.

A more realistic freshwater reservoir-management benchmark is being developed to investigate resource-allocation conflicts among heterogeneous agents interacting through shared hydrological dynamics. Longer-term extensions aim to support additional optimization algorithms and dynamically adapting regulatory agents.

Although motivated by renewable resource management, the framework is applicable to a broader class of problems in which adaptive agents interact under modifiable institutional rules. Potential application domains include fisheries, water resource allocation, environmental regulation, transportation systems, and other socio-ecological systems involving strategic interactions.

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By separating reinforcement learning algorithms from mechanism optimization, the framework enables researchers to investigate new classes of adaptive governance problems while leveraging existing reinforcement learning ecosystems.

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AI usage disclosure

Generative AI tools, including OpenAI ChatGPT (GPT-6), were used to assist with drafting documentation and code review. The human authors are responsible for the framework's research objectives, architectural decisions, scientific methodology, and final implementation. All AI-assisted materials were reviewed and edited by the authors, and the software was validated using the project's automated testing and reproducible experiment workflows.

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